Spatiotemporal Assessment of Particulate Matter in an Organized Industrial Zone

Using Low-Cost Sensor Networks: A Case Study from İnegöl, Türkiye

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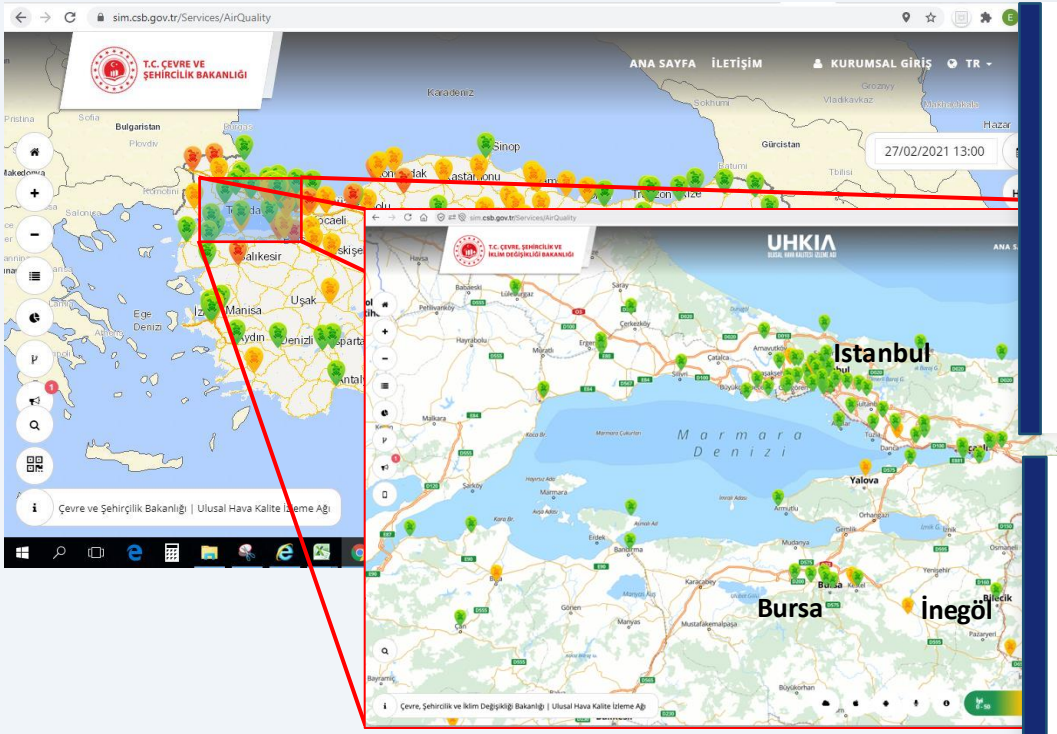
Monitoring · Data Quality · Source Attribution · Exposure Assessment

Air Quality in Türkiye – The National Picture



Türkiye's national PM10 limit: 40 ug/m3 annual mean – aligned with previous EU standard.

No national annual limit exists for PM2.5.

Monitoring network is sparse, regulatory stations are concentrated in major metropolitan areas, leaving mid-sized industrial cities largely unobserved.

Bursa province is one of Türklye's most industrialized regions – automotive, textile and furniture manufacturing.

İnegöl sits within Bursa specialized furniture and production hub, surrounded by 'Organized Industrial Zones (OIZs).

Why Inegol ? — Motivation & Context

What We Already Know

PM_{10} 75.8	µg/m³ recorded in İnegöl in 2017; slight downward trend over 2014–2017 (Çalışkan et al., 2025)
30% PAH	in particulate phase — above typical urban range; attributed to furniture sector emissions
BTEX 34.7	µg/m³ average — benzene, toluene & xylenes from wood processing and coatings (Çalışkan et al., 2022)
Gap	No spatially dense PM network or OIZ source-attribution study had been conducted before this work

Study Objectives

① Quantify PM_{10} & $PM_{2.5}$

Across 10 strategically selected sites over a full year (2021–2022)

② Evaluate OIZ Influence

Assess organized industrial zones' contribution to local & regional PM levels

③ Source–Receptor Analysis

Identify pollution hotspots and quantify increments above background

④ Wind & Morphology Effects

Analyse how meteorological conditions drive spatiotemporal PM variability

Monitoring Network & Site Classification

Network Highlights

10 sites	across İnegöl district
Airqoon®	IoT-enabled, solar-powered sensors
Continuous	PM_{10}, $PM_{2.5}$, temp, humidity
2021–2022	one full year of monitoring

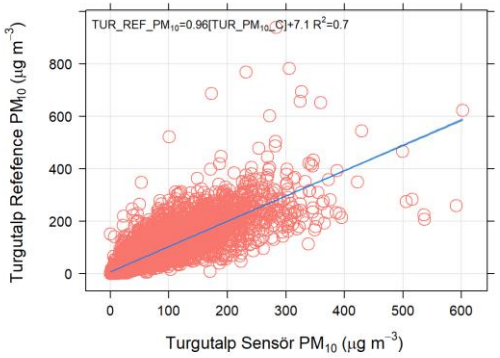
Sensor Calibration & Data Quality

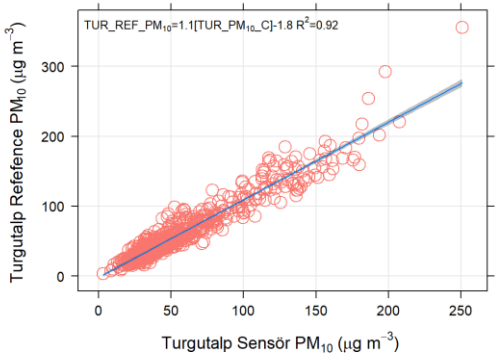
Calibration Approach

Co-location at the Turgutalp reference station (Ministry of Environment — MEUCC)

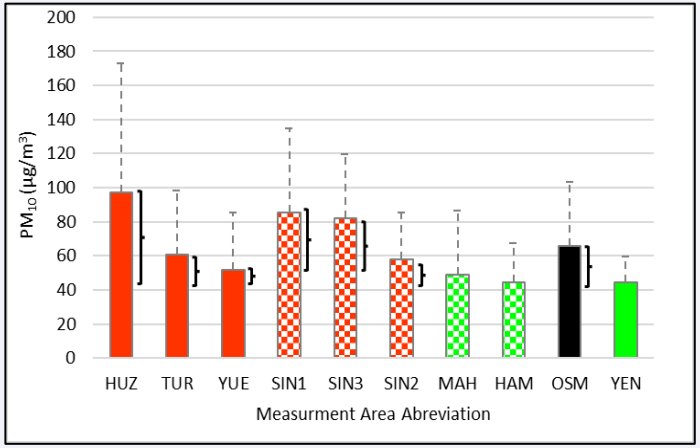
Field calibration against BAM regulatory monitors for PM_{10}

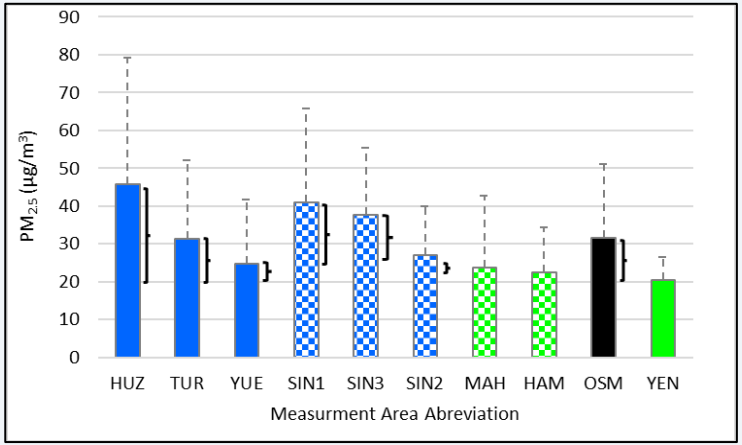
Hourly PM10 r2-score: 0.84, daily PM10 r2-score: 0.96, Daily MAE: 4.5 µg/m³





Annual Mean PM Concentrations





96.6 µg/m³

HUZ – highest urban PM_{10}
(solid fuel heating)

85.2 µg/m³

SIN1 – highest industrial PM_{10}
(metal + furniture + textile)

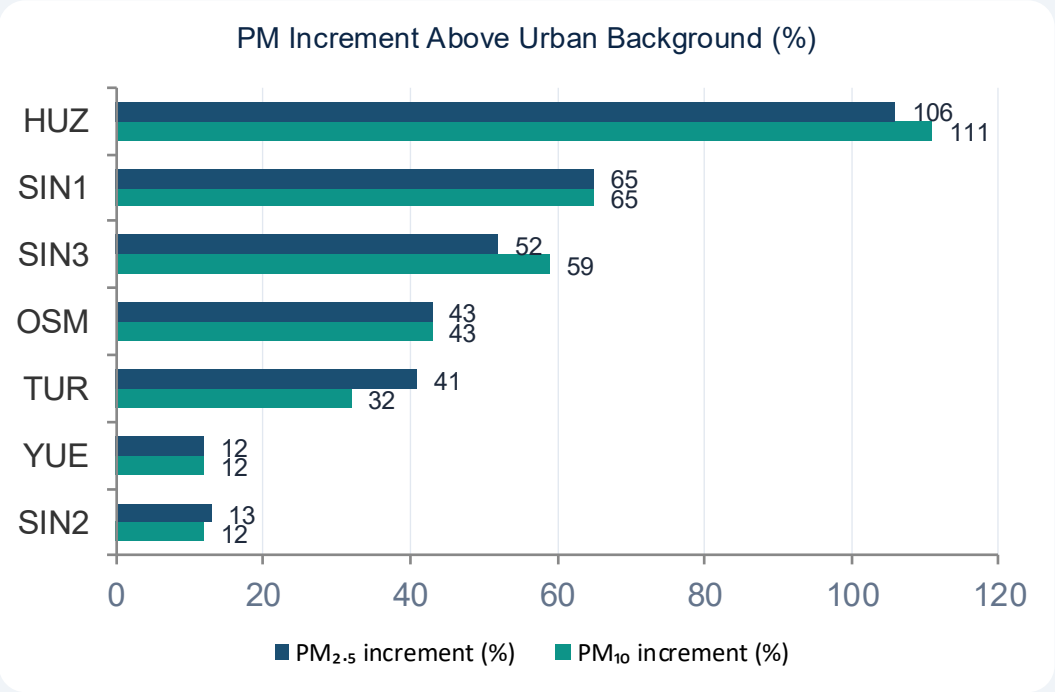
5–11×

WHO $PM_{2.5}$ guideline (5 µg/m³)
exceeded across ALL sites

100%

Sites exceed national
HKDYY PM_{10} limit (40 µg/m³)

Source Increment Analysis



Residential Heating (HUZ)

+111% PM_{10} | +106% $PM_{2.5}$
Solid fuel use in low-income area

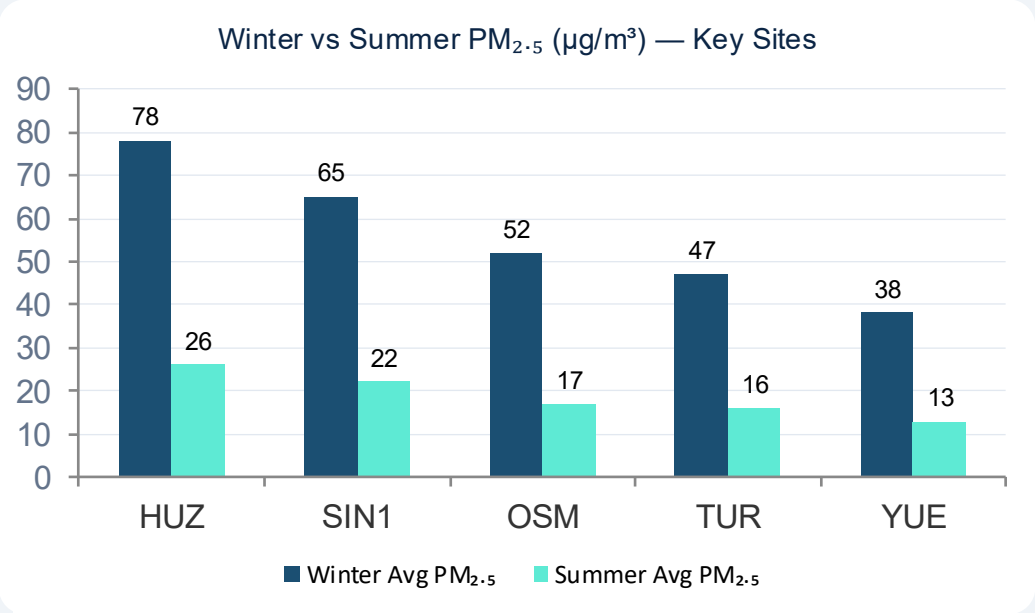
Industrial OIZ (SIN1)

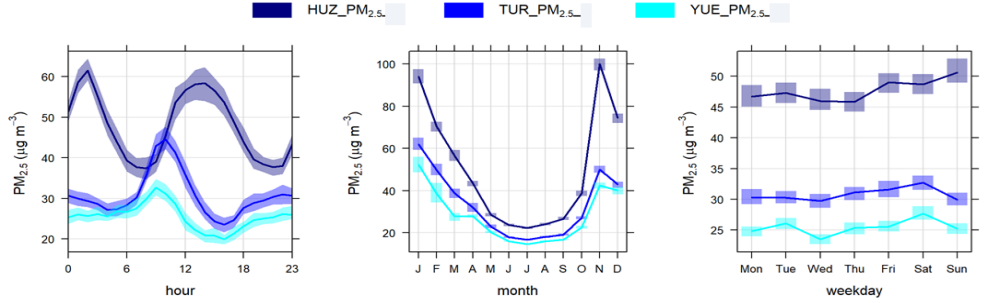
+65% PM_{10} | +65% $PM_{2.5}$
Metal, textile, furniture sector

Traffic (OSM)

+43% PM_{10} | +43% $PM_{2.5}$
Exceeds Istanbul, Berlin & London

Seasonal & Diurnal Patterns





Winter 2–3× Summer

Residential heating & secondary aerosol formation drive cold-season peaks

Traffic Peaks at OSM

Morning (08:00–09:00) & evening rush hours clearly captured

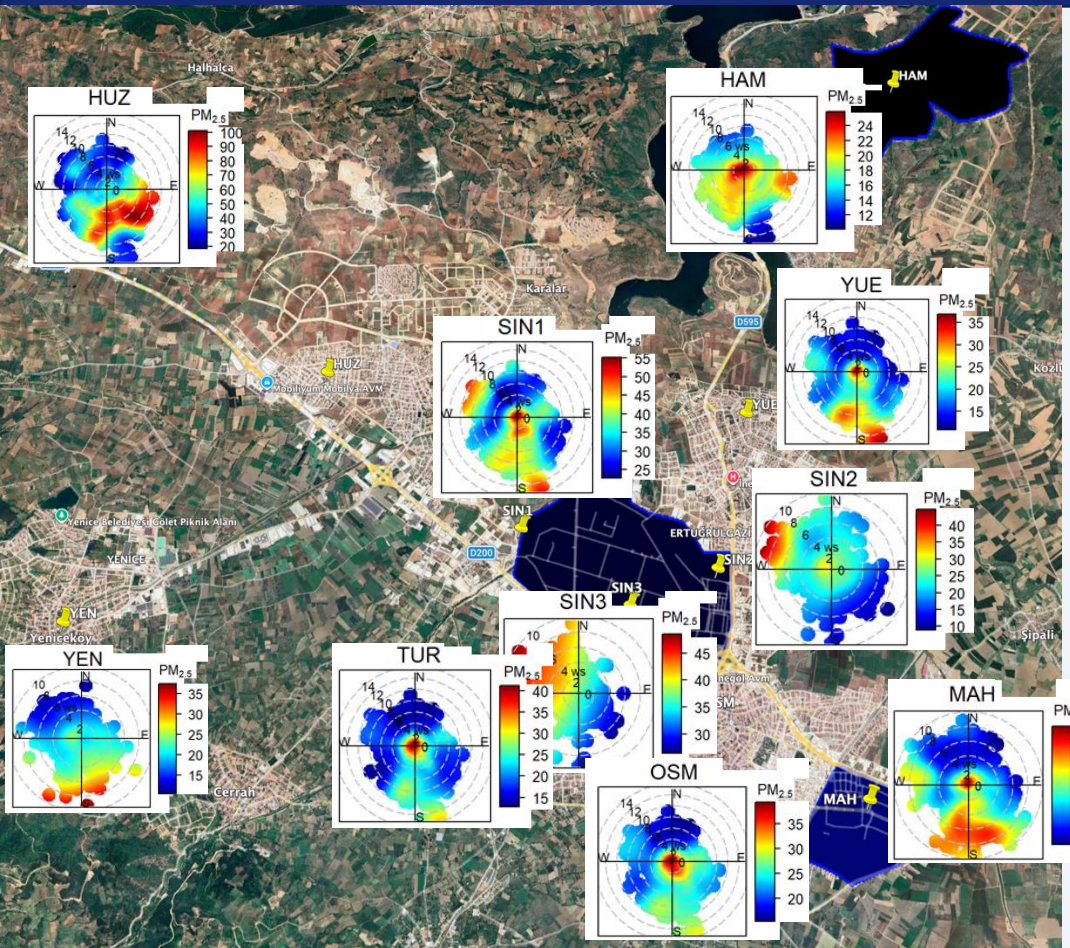
HUZ — Dual Daily Peak

Midday secondary aerosol + nighttime solid fuel combustion peak

Lowest on Sundays

Except HUZ & HAM — residential heating peaks on weekends

Wind–$PM_{2.5}$ Coupling



South winds → elevated $PM_{2.5}$

Dominant pattern at most sites. Transports emissions from Sinanbey industrial zone southward into residential areas.

West winds → SIN2, SIN3 peaks

Sinanbey industrial complex lies entirely to the west flow concentrates metal & furniture emissions at these sensors.

Calm winds = highest local loading

Dense circular patterns in polar plots near OSM & HAM indicate stagnant air causes pollutant trapping under low-dispersion conditions.

Spatial Heterogeneity Analysis

R = 0.41

Pearson correlation between sites

vs. Istanbul R = 0.72

COD < 0.15

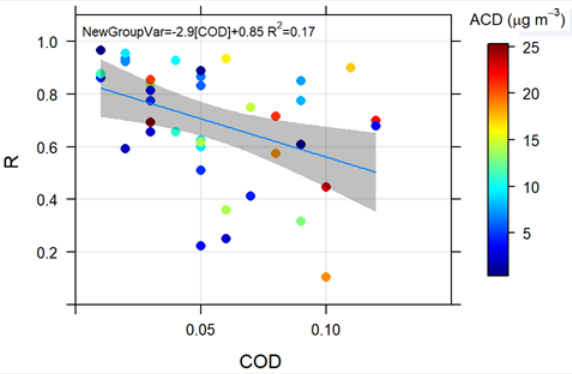
Coefficient of Divergence (spatial uniformity index)

Local sources dominate

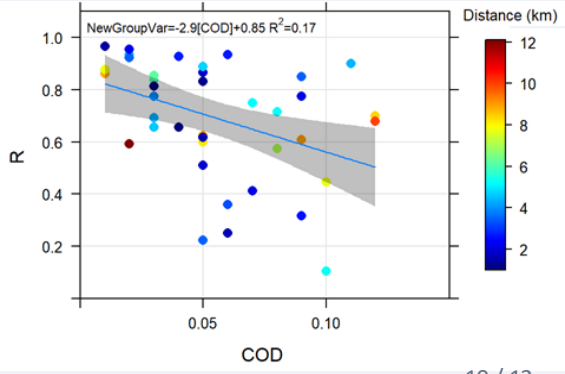
What This Tells Us

- Weaker inter-site correlation (R = 0.41) than Istanbul (R = 0.72) confirms more complex local PM variability in İnegöl.
- Spatial heterogeneity is driven by local emission sources — industrial zones, traffic, and solid fuel heating — not regional background transport.
- Unlike São Paulo, Beijing, or Augsburg, inter-site distance is NOT a predictor of PM variability in İnegöl — local morphology dominates.

Influence of Inter-Station Distance



Influence of Absolute Concentration Difference



Key Takeaways & Policy Implications

Industrial OIZs are dominant pollution drivers

Sinanbey (SIN1) recorded 65% above background PM. Industrial siting and emission controls need urgent review.

Dense LCAQS networks fill monitoring gaps

Low-cost sensors effectively resolve fine-scale pollution gradients that sparse official networks miss.

Solid fuel heating creates extreme winter peaks

HUZ concentrations reach 3× summer levels. Transitioning low-income areas to clean energy is a public health priority.

Urban morphology governs exposure

Building layout and wind corridors create stagnation zones with up to 2× elevated concentrations.

Traffic emissions exceed international benchmarks

$PM_{2.5}$ increment at OSM (10 µg/m³) surpasses Istanbul, Berlin, London. Urban traffic management is critical.

Conclusions & Future Directions

Key Conclusions

- All sites exceed national (HKDYY) and international (WHO) PM limits
- Industrial OIZs, solid fuel heating & traffic are the three dominant source classes
- Winter-to-summer ratio of 2–3× confirms seasonal combustion as a major driver
- Spatial heterogeneity is locally controlled — not distance-dependent
- Dense LCAQS networks are a scalable tool for industrial zone monitoring

Future Research

- Multi-height sensor deployments in street canyons & courtyards
- Real-time meteorological coupling & wet/dry deposition modeling
- Indoor–outdoor PM infiltration monitoring for exposure assessment
- Comparative analysis across Turkish industrial mid-sized cities
- Linking PM data to health outcomes for evidence-based policy

Thank you!

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